

Roots of Random Polynomials over a Finite Field

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Suppose that F_q is a finite field and $F_q[x]$ is the ring of polynomials over the field F_q . For a polynomial $f(x) \in F_q[x]$, we denote by $\lambda(f)$ the number of its roots lying in the field F_q . In this paper, we discuss the problem of the statistical properties of $\lambda(f)$, provided that the normalized polynomials¹ of degree at most m are uniformly distributed on the set. Such problems occur in the study of permutation polynomials and in the theory of error-correcting codes [1]–[3]. For a real field $\mathbb{R}$, Kac's theorem asserts that, “on the average,” a polynomial of degree n with real coefficients normally and independently distributed on the closed interval $[0, 1]$ has, asymptotically, $(\log_2 n)/(2\pi)$ real roots [4].

The main result of this paper is as follows: the number of zeros of a random polynomial from $F_q[x]$ lying inside the field F_q has, asymptotically as $(q \rightarrow \infty)$, a Poisson distribution with parameter $\lambda = 1$. In particular, a random polynomial from $F_q[x]$ has “on the average,” as q increases, exactly one root in the field F_q .

Let t_m be the number of normalized (with leading coefficient one) polynomials of degree m having no roots in the field F_q (here $q = p^r$ and p is a prime).

Lemma 1. *The following relation holds:*

$$t_m = \sum_{i=0}^m (-1)^i \binom{q}{i} q^{m-i}. \quad (1)$$

Proof. Let Z_m^i denote the set of polynomials of degree m from $F_q[x]$ having a root of the form $\alpha_i \in F_q = \{\alpha_1 = 0, \alpha_2, \dots, \alpha_q\}$, and let

$$Z_m = \bigcup_{i=1}^q Z_m^i.$$

Then

$$t_m = q^m - |Z_m|, \quad (2)$$

where $|Z_m|$ is the cardinality of the set Z_m . Applying the “inclusion–exclusion” formula, from (2) we obtain

$$|Z_m| = \sum_{s=1}^m (-1)^{s-1} \sum_{1 \leq i_1 \leq \dots \leq i_s \leq q} |Z_m^{i_1} \cap Z_m^{i_2} \cap \dots \cap Z_m^{i_s}|. \quad (3)$$

¹In what follows, only normalized polynomials from $F_q[x]$ are considered unless stated otherwise.

If a normalized polynomial $f(x) \in F_q[x]$ of degree m has, for roots, elements $\alpha_{i_1}, \alpha_{i_2}, \dots, \alpha_{i_s}$ from the field F_q , then

$$f(x) = (x - \alpha_{i_1}) \dots (x - \alpha_{i_s}) \varphi(x),$$

where $\varphi(x)$ is an arbitrary normalized polynomial of degree $m - s$ from $F_q[x]$. Hence

$$|Z_m^{i_1} \cap Z_m^{i_2} \cap \dots \cap Z_m^{i_s}| = q^{m-s}. \quad (4)$$

Relations (2)–(4) imply (1). $\square$

Applying the lemma, let us calculate, for example, the values of t_m for $m = 0, 1, 2, 3$: $t_0 = 1$, $t_1 = 0$, $t_2 = \binom{q}{2}$, $t_3 = q(q^2 - 1)/3$.

Corollary 1. *Suppose that $B_{m,q,r}$ is the number of polynomials of degree m from $F_q[x]$ having exactly r different roots in the field F_q . Then*

$$B_{m,q,r} = \binom{q}{r} \sum_{i=0}^{m-r} \binom{r+i-1}{r-1} t_{m-(r+i)}. \quad (5)$$

Proof. If the polynomial $f(x)$ has r different roots in the field F_q , then

$$f(x) = (x - \alpha_1)^{\lambda_1} (x - \alpha_2)^{\lambda_2} \dots (x - \alpha_r)^{\lambda_r} \varphi(x),$$

where all the α_i are different, $\lambda_i \geq 1$, and $\varphi(x)$ has no roots in F_q . This yields the relation

$$B_{m,q,r} = \binom{q}{r} \sum_{k=r}^m \binom{k-1}{r-1} t_{m-k} = \binom{q}{r} \sum_{i=0}^{m-r} \binom{r+i-1}{r-1} t_{m-(r+i)}. \quad \square$$

Further, suppose that

$$t_{m,\lambda}(q) = \sum_{i=0}^m (-1)^i \binom{q}{i} \lambda^{m-i}$$

and $F(x)$ is the generating function of the sequence $\{t_{m,\lambda}(q)\}_{m=0}^{\infty}$, i.e.,

$$F(x) = \sum_{m=0}^{\infty} t_{m,\lambda}(q) x^m.$$

Lemma 2. *The following relation holds:*

$$F(x) = \frac{(1-x)^q}{1-\lambda x}. \quad (6)$$

Relation (6) is proved by a straightforward calculation.

Further, we use the standard formalism from [5]:

$$\begin{aligned} \binom{r+i-1}{i} &= \text{Coef}_u \left\{ \frac{1}{(1-u)^r u^{i+1}} \right\}, \\ t_{m-(r+i)} &= \text{Coef}_u \left\{ \frac{(1-u)^q}{(1-uu) u^{m-r-i+1}} \right\}, \\ t_{m,\lambda}(q) &= \text{Coef}_u \left\{ \frac{(1-u)^q}{(1-\lambda u) u^{m+1}} \right\}. \end{aligned} \quad (7)$$

Lemma 3. *The following relation holds:*

$$B_{m,q,r} = \binom{q}{r} t_{m-r,q}(q-r). \quad (8)$$

Proof. From (5) and (7), we find

$$\sum_{i=0}^{m-r} \binom{r+i-1}{r-1} t_{m-(r+i)} = \sum_{i=0}^{\infty} \text{Coef}_u \left\{ \frac{1}{(1-u)^{i+1} u^r} \right\} \text{Coef}_v \left\{ \frac{(1-v)^q}{(1-vq)v^{m-r-i+1}} \right\}. \quad (9)$$

From (9), after straightforward calculations, we obtain

$$\begin{aligned} \sum_{i=0}^{m-r} \binom{r+i-1}{r-1} t_{m-(r+i)} &= \text{Coef}_v \left\{ \frac{(1-v)^q}{(1-vq)v^{m-r+1}} \text{Coef}_u \left\{ \frac{1}{u^r(1-u)} \cdot \frac{1}{1-\frac{v}{1-u}} \right\} \right\} \\ &= \text{Coef}_v \left\{ \frac{(1-v)^q}{(1-vq)v^{m-r+1}(1-v)^r} \right\} = t_{m-r,q}(q-r). \quad \square \end{aligned}$$

Suppose that $N_m = \{f_1, f_2, \dots, f_{q^m}\}$ is the set of all polynomials of degree m from $F_q[x]$. On N_m , let us introduce a uniform probability distribution by assigning to each polynomial from N_m the probability $1/q^m$. Consider the random variable $\lambda(f)$, which is the number of different roots of the polynomial f lying in the field F_q , where f is a randomly chosen polynomial from N_m . Consider the distribution

$$q_r^{(m)} = \mathbf{P}\{\lambda(f) = r\}, \quad r = 0, 1, 2, \dots,$$

and the generating function of this distribution

$$\Phi_m^q(x) = \sum_{r=0}^{\infty} q_r^{(m)} x^r.$$

Lemma 4. *The following relation holds:*

$$\Phi_m^q(x) = \sum_{i=0}^m \frac{\binom{q}{i}}{q^i} (x-1)^i. \quad (10)$$

Proof. Using Lemma 3, we obtain

$$\begin{aligned} \Phi_m^q(x) &= \sum_{r=0}^m \frac{B_{m,q,r}}{q^m} x^r = \frac{1}{q^m} \sum_{r=0}^m x^r \binom{q}{r} t_{m-r,q}(q-r) \\ &= \frac{1}{q^m} \sum_{r=0}^m x^r \binom{q}{r} \text{Coef}_u \left\{ \frac{(1-u)^{q-r}}{(1-uq)u^{m-r+1}} \right\} \\ &= \frac{1}{q^m} \text{Coef}_u \left\{ \frac{(1-u)^q}{u^{m+1}(1-uq)} \sum_{r=0}^{\infty} \left(\frac{ux}{1-u} \right)^r \binom{q}{r} \right\} = \frac{1}{q^m} \text{Coef}_u \left\{ \frac{(1+u(x-1))^q}{(1-uq)u^{m+1}} \right\} \\ &= \frac{1}{q^m} \sum_{s=0}^q \binom{q}{s} (x-1)^s \text{Coef}_u \left\{ \frac{1}{u^{m-s+1}} [1 + (uq) + (uq)^2 + \dots] \right\} \\ &= \frac{1}{q^m} \sum_{s=0}^q \binom{q}{s} (x-1)^s q^{m-s} = \sum_{s=0}^m \frac{\binom{q}{s}}{q^s} (x-1)^s. \end{aligned}$$

In all the calculations, the inequality $r \leq \min\{q, m\}$ is taken into account. $\square$

Next, let $M(\lambda(f))$ and $D(\lambda(f))$ be, respectively, the expectation and variance of the random variable $\lambda(f)$.

Theorem 1. *The following relations hold:*

$$M(\lambda(f)) = 1, \quad D(\lambda(f)) = 1 - \frac{1}{q}. \quad (11)$$

This assertion immediately follows from Lemma 4, because the first and second derivatives of the function $\Phi_m^q(x)$ are easily calculated at the point $x = 1$.

Next, suppose that $M_n = \{f\}$ is the set of all polynomials of degree $\leq n$ from $F_q[x]$. On the set M_n , let us introduce a uniform probability distribution by assigning to each polynomial f from M_n the probability $1/N$, where $N = |M_n| = (q^{n+1} - 1)/(q - 1)$. Consider the same random variable $\lambda_n(f)$, which is the number of different roots of the polynomial f from M_n lying in the field F_q .

Theorem 2. *If $n = q - 1$ and $q \rightarrow \infty$, then the random variable $\lambda_n(f)$ has, in the limit, a Poisson distribution with parameter $\lambda = 1$.*

Proof. Using Lemma 3 and the rule of total probability, we obtain

$$\begin{aligned} P\{\lambda_n(f) = r\} &= P_r = \frac{1}{N} \sum_{m=1}^n q^m q_r^{(m)} = \frac{\binom{q}{r}}{N} \sum_{m=1}^n \text{Coef}_u \left\{ \frac{(1-u)^{q-r}}{(1-qu)u^{m-r+1}} \right\} \\ &= \frac{\binom{q}{r}}{N} \text{Coef}_u \left\{ \frac{(1-u)^{q-r-1}}{(1-qu)u^{n-r+1}} \right\} = \frac{\binom{q}{r}}{N} t_{n-r,q}(q-r-1). \end{aligned} \quad (12)$$

With the help of (12), we can find the generating function of the distribution $\{P_r\}$:

$$\begin{aligned} \Phi_q^n(x) &= \frac{1}{N} \sum_{r=0}^{\infty} \binom{q}{r} t_{n-r,q}(q-r-1) x^r = \frac{1}{N} \sum_{r=0}^{\infty} \binom{q}{r} x^r \text{Coef}_u \left\{ \frac{(1-u)^{q-r-1}}{(1-qu)u^{n-r+1}} \right\} \\ &= \frac{1}{N} \text{Coef}_u \left\{ \frac{(1-u)^{q-1}}{(1-qu)u^{n+1}} \sum_{r=0}^q \binom{q}{r} \left(\frac{xu}{1-u} \right)^r \right\} = \frac{1}{N} \text{Coef}_u \left\{ \frac{[1+u(x-1)]^q}{(1-qu)(1-u)u^{n+1}} \right\}. \end{aligned}$$

Hence

$$\Phi_q^n(x) = \frac{1}{2\pi i N} \oint_{|u|=\rho} \frac{[1+u(x-1)]^q}{(1-qu)(1-u)u^{n+1}} du. \quad (13)$$

Here ρ is a sufficiently small positive number.

The function $\Psi(u)$ defined as

$$\Psi(u) = \frac{[1+u(x-1)]^q}{(1-uv)(1-u)u^{n+1}}$$

has the following singular points: $u_1 = 0$, $u_2 = q^{-1}$, $u_3 = 1$, $u_4 = \infty$. Since $n = q - 1$, it follows that

$$\text{res}_{u=u_4} \Psi(u) = 0.$$

Further,

$$\text{res}_{u=u_2} \Psi(u) = q^n \frac{(1+(x-1)/q)^q}{1/q-1}, \quad \text{res}_{u=u_3} \Psi(u) = \frac{x^q}{q-1}.$$

Now, using the residue theorem, we obtain

$$\Phi_q^n(x) = \frac{q^{n+1} - q^n}{q^{n+1} - 1} \left(1 - \frac{x-1}{q} \right)^q - \frac{x^q}{q^{n+1} - 1}. \quad (14)$$

It follows from (14) that

$$\lim_{q \rightarrow \infty} \Phi_q^n(x) = e^{x-1}$$

for an arbitrary point x from $(0, 1]$. The assertion of Theorem 2 now follows from the continuity theorem for the generating function [6]. $\square$

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BIBLIOGRAPHY

1. R. Lidl and H. Niederreiter, *Finite Fields*, Addison-Wesley Publishing Co., Reading, Mass., 1983; Russian transl.: vols. 1 and 2, Mir, Moscow, 1988..
2. F. J. MacWilliams and N. J. A. Sloane, *The Theory of Error-Correcting Codes*, North-Holland Publishing Co., Amsterdam–New York–Oxford, 1977; Russian transl.: Svyaz', Moscow, 1979..
3. V. K. Leont'ev, *Proc. IV Internat. Conference "Discrete Models in Control Systems Theory"*, 2000.
4. M. Kac, *Probability and Related Topics in Physical Sciences*, Interscience, London, 1959; Russian transl.: Mir, Moscow, 1965.
5. G. P. Egorychev, *Integral Representation and Calculation of Combinatorial Sums* [in Russian], Nauka, Novosibirsk, 1977.
6. V. N. Sachkov, *Probability Methods in Combinatorial Analysis* [in Russian], Nauka, Moscow, 1978.

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